Commutative Algebra Problem Set #3

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Problem (Exercise 1)**.** *Let $A = \mathbb{Z}$ and $M = \mathbb{Q}/\mathbb{Z}$. Show that $M(x) = M \otimes_A \kappa(x)$ is zero for every $x \in Spec(A)$.*

Solution. First, consider $x = 0$ separately:

$$M(0) = M \otimes_{\mathbb{Z}} \kappa(0) = M \otimes_{\mathbb{Z}} \mathrm{Frac}(\mathbb{Z}) \cong (\mathbb{Q}/\mathbb{Z}) \otimes_{\mathbb{Z}} \mathbb{Q} \cong 0$$

The last isomorphism comes from the fact that for any $\frac{n}{m}+\mathbb{Z} \in \mathbb{Q}/\mathbb{Z}$ and $\frac{p}{q} \in \mathbb{Q}$:

$$\left(\frac{n}{m}+\mathbb{Z}\right) \otimes \frac{p}{q} = \left(\frac{n}{m}+\mathbb{Z}\right) \otimes m \cdot \frac{p}{qm} = m\left(\frac{n}{m}+\mathbb{Z}\right) \otimes \frac{p}{qm} = 0.$$

Now consider any non-zero prime ideal $p\mathbb{Z}$ corresponding to x. Then, $\kappa(x) = \mathrm{Frac}(A/p\mathbb{Z}) = \mathrm{Frac}(\mathbb{Z}/p\mathbb{Z}) \cong \mathrm{Frac}(\mathbb{F}_p) \cong \mathbb{F}_p$. Thus we consider:

$$M(x) = (\mathbb{Q}/\mathbb{Z}) \otimes_{\mathbb{Z}} \mathbb{F}_p.$$

This is again clearly 0. For arbitrary $\frac{n}{m} + \mathbb{Z} \in \mathbb{Q}/\mathbb{Z}$ and $\alpha \in \mathbb{F}_p$ we have:

$$\left(\frac{n}{m}+\mathbb{Z}\right) \otimes \alpha = p\left(\frac{n}{mp}+\mathbb{Z}\right) \otimes \alpha = \left(\frac{n}{mp}+\mathbb{Z}\right) \otimes p\alpha = 0.$$

Problem (Exercise 2)**.** *Let A be a ring and M a finitely presented A-module: that is, exists a presentation*

$$A^m \xrightarrow{\alpha} A^n \xrightarrow{\pi} M \longrightarrow 0,$$

*or equivalently there exists a surjection $A^n \to M$ whose kernel is finitely generated. Show that the kernel of **every** surjection $A^k \to M$ is finitely generated.*

Proof. As noted in the hint, we first show that if

$$0 \to M' \xrightarrow{\iota} M \xrightarrow{\pi} M'' \to 0$$

is exact and both M' and M'' are finitely generated, then so is M. Let M' be generated by $x_1, \ldots, x_n$ while M'' by $y_1, \ldots, y_m$. Let $z_1, \ldots, z_m$ be pre-images of $y_1, \ldots, y_m$ in M (that is, $\pi(y_i) = z_i$). We claim $M = \langle x_1, \ldots, x_n, z_1, \ldots, z_m \rangle$, and so M is finitely generated.

Indeed, take any $v \in M$. Note that $\pi(v) = \sum_{i=1}^{m} \alpha_i y_i$ for $\alpha_i \in A$ since M'' is finitely generated. Thus trivially, $v - \sum_{i=1}^{m} \alpha_i z_i \in \ker(\pi) \cong \mathrm{im}(\iota)$, and so we can write this difference using generators of M':

$$v - \sum_{i=1}^{m} \alpha_i z_i = \sum_{j=1}^{n} \beta_i x_i, \quad \beta_i \in A$$

Thus, we have found the linear combination $\sum_{i=1}^{m} \alpha_i z_i + \sum_{i=1}^{n} \beta_i x_i$ for any $v \in M$.

Now, we shall show that **every** surjection $A^k \to M$ is finitely generated. Build the following commutative diagram:

$$\begin{array}{ccccccccc} & & A^m & \longrightarrow & A^n & \longrightarrow & M & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle g} & & \downarrow{\scriptstyle f} & & \| & & \\ 0 & \longrightarrow & K & \longrightarrow & A^k & \xrightarrow{\pi} & M & \longrightarrow & 0 \end{array}$$

Here, $f : A^n \to A^k$ is constructed by lifting the identity map id_M, and the map $g : A^m \to K$ is obtained by restricting the map f. Now apply the Snake's lemma to get exact sequence:

$$0 \to \mathrm{coker}(g) \to \mathrm{coker}(f) \to 0.$$

Thus, $\mathrm{coker}(f) \cong \mathrm{coker}(g)$, and so we have the exact sequence:

$$0 \to \mathrm{im}(g) \to K \to \mathrm{coker}(f) \to 0,$$

where both $\mathrm{im}(g)$ and $\mathrm{cokeg}(f)$ are finitely generated. Thus, K is finitely generated as well. □

Problem (Exercise 3)**.** *Let A be a domain. Recall that an A-module M is* ***torsion-free*** *if for every nonzero $x \in A$, the map $\mu_x : m \mapsto xm$ is injective.*

(a) Show that every flat A-module is torsion-free.

(b) Suppose that A is a PID. Show that every torsion-free A-module is flat.

(c) Give example of a domain A and torsion-free A-module which is not flat.

Solution.

Part (a). Consider the exact sequence:

$$0 \to A \xrightarrow{\mu_x} A.$$

Let us apply the $(\cdot) \otimes_A M$ functor (which leaves sequence exact since A is flat):

$$0 \to M \xrightarrow{\mu_x} M.$$

Thus $\mu_x : M \to M, m \mapsto xm$ is injective, and so M is torsion-free. □

Part (b). As noted in the hint, assume first that M is a finitely generated A-module with A being a PID. Due to the Structure Theorem, $M \cong A^r \oplus T$ with $T = \bigoplus_{i=1}^{m}(A/a_i)$ for $a_1 \mid \cdots \mid a_m$. In particular, when M is torsion-free, M is free, so $M \cong A^r$. But free modules are always flat, so this part is finished.

Now consider the general case, where M is not necessarilly finitely generated. However, M is a direct limit of finitely generated submodules: say, $M = \varinjlim M_\alpha$. Each M_α is clearly torsion-free, so from the first half, we know that each M_α is flat. Now, informally, we are left to show that the filtered colimit of flat modules is, again, a flat module.

Consider an exact sequence $0 \to N' \to N \to N'' \to 0$. Since each M_α is flat, we get an exact sequence $0 \to N' \otimes_A M_\alpha \to N \otimes_A M_\alpha \to N'' \otimes_A M_\alpha \to 0$. Then apply the filtered colimit to this sequence. Note two things: (1) tensor product commutes with filtered colimits, so $N \otimes_A M \cong \varinjlim(N \otimes_A M_\alpha)$, (2) colimit functor is exact. Combining these two facts, we get an exact sequence

$$0 \to N' \otimes_A M \to N \otimes_A M \to N'' \otimes_A M \to 0,$$

so $(\cdot) \otimes_A M$ functor is exact, so M is flat. □

Part (c). Take $A := k[x, y]$ and $M := (x, y)$. Clearly, A is a domain and M is torsion-free. However, M is not flat. Indeed, if that was the case, for every finitely generated ideal $I \subseteq A$, the map $I \otimes_A M \hookrightarrow M$ must be injective*. However, take $I = (x, y)$. Then, $\gamma : (x, y) \otimes (x, y) \to (x, y), f \otimes g \mapsto fg$ is not injective. Indeed, notice that $0 \neq x \otimes y - y \otimes x \in \ker(\gamma)$.

Problem (Exercise 4)**.** *Find an example of a homomorphism between domains $A \to B$ and a torsion-free A-module M such that $M \otimes_A B$ is not torsion-free.*

Solution. Let $A = k[x, y]$ and $B = k[x]$, where the homomorphism $k[x, y] \to k[x]$ is given by $f(x, y) \mapsto f(x, 0)$. Let $M = (x, y)$. Let us show that $M \otimes_A B$ is not torsion-free. We have:

$$\begin{aligned} M \otimes_A B &= (x, y) \otimes_{k[x,y]} k[x] \\ &\cong (x, y) \otimes_{k[x,y]} (k[x, y]/(y)) \\ &\cong (x, y)/((x, y) \cdot (y)) \\ &\cong (x, y)/(xy, y^2) \end{aligned}$$

However, this quotient is clearly not torsion-free. Indeed, take the element $z := y \in M \otimes_A B$ and $x \in B$. Then, $zx = 0$ in $M \otimes_A B$, so the multiplication-by-x map is not injective in $M \otimes_A B$.

*https://stacks.math.columbia.edu/tag/00H9, Lemma 10.39.5(3)

Problem (Exercise 5)**.** *This problem consists of two parts:*

(a) Let I be a finitely generated ideal in a local ring $(A, \mathfrak{m})$. Prove that if $I = I^2$, then $I = A$ or $I = 0$.

(b) Let $A = C(\mathbb{R}, \mathbb{R})$ be the ring of continuous functions $\mathbb{R} \to \mathbb{R}$ and $\mathfrak{m} = \{f \in A : f(0) = 0\}$. Prove that $\mathfrak{m} = \mathfrak{m}^2$ and conclude that the ideal $\mathfrak{m}$ is not finitely generated.

Solution.

Part (a). Suppose $I \subsetneq \mathfrak{m}$ (otherwise $I = A$, and so we are done). Then, $I = I^2 \subseteq \mathfrak{m}I$. Apply Nakayama's lemma to get $I = 0$.

Alternatively, we can show this directly: suppose $I = (x_1, \dots, x_n)$. Since $I \subseteq \mathfrak{m}I$, $x_i = \sum_{j=1}^{n} m_{ij} x_j$ for $m_{ij} \in \mathfrak{m}$. Then, we have $\mathbf{x} = M\mathbf{x}$ for $\mathbf{x} \in A^n$ consisting of $x_1, \dots, x_n$. Note that the matrix $T := E_{n\times n} - M$ is invertible. Indeed, consider the natural projection map $\pi : A \to A/\mathfrak{m}$, and so $\pi(T) = \pi(E_{n\times n}) = E_{n\times n}$. Thus, $\det(T) = 1$ in $A/\mathfrak{m}$, so in particular the determinant is not in $\mathfrak{m}$, and so $\det(T)$ is a unit in A. This implies that T is invertible, so the equation $T\mathbf{x} = 0$ has the only solution $\mathbf{x} = 0$. Thus, $I = 0$. □

Part (b). Clearly, $\mathfrak{m}^2 \subseteq \mathfrak{m}$, since $I^2 \subseteq I$ holds for any ideal I. For $\mathfrak{m} \subseteq \mathfrak{m}^2$, take $f \in C(\mathbb{R}, \mathbb{R})$ with $f(0) = 0$. Choose

$$g(x) := \operatorname{sign}(f(x))\sqrt{|f(x)|}, \quad h(x) := \sqrt{|f(x)|}$$

Notice that $g \in \mathfrak{m}$ (this function is continuous and $g(0) = 0$), and similarly $h \in \mathfrak{m}$. Moreover, $f = gh$. Thus $\mathfrak{m}^2 = \mathfrak{m}$.

Now suppose $\mathfrak{m}$ is finitely generated. Consider the localization $A_{\mathfrak{m}}$. Since localization preserves finite generation, either $\mathfrak{m}A_{\mathfrak{m}} = 0$ or $\mathfrak{m}A_{\mathfrak{m}} = A_{\mathfrak{m}}$. The latter is impossible, so $\mathfrak{m}A_{\mathfrak{m}} = 0$. This means that there exists some continuous f such that $f(0) \neq 0$ and $f \cdot g = 0$ for every $g \in \mathfrak{m}$. This is clearly impossible. Denote by $B_\varepsilon(x_0) = \{x \in \mathbb{R} : |x - x_0| < \varepsilon\}$. Then,

(1) there exists a neighbourhood $B_\varepsilon(0)$ such that $B_\varepsilon(0) \subset \operatorname{supp}(f)$;

(2) we can pick $g \in A$ such that $\operatorname{supp}(g) = B_{|x_0|/2}(x_0) \not\ni 0$ for $x_0 \in B_{\varepsilon/2}(0)$[†].

Then, $g(0) = 0$, so $g \in \mathfrak{m}$. Moreover, $f(x_0)g(x_0) \neq 0$, and so $fg \neq 0$. □

[†]For instance, take a bump function: https://ncatlab.org/nlab/show/bump+function.